

ECN 594: Midterm Exam

February 9, 2026

Instructions:

- You have **80 minutes**
- You may bring a calculator and notes on a two-sided cheat-sheet (letter-size paper)
- Please be neat. If your work is too messy it will not be graded.
- Be sure to show your working.
- This is a long exam, so there are lots of ways to get points. If you get stuck, move on!
- Good luck!

Name: _____

Question:	1	2	3	4	Total
Points:	30	30	20	20	100
Score:					

1. Short Answer Questions (30 points)

1. For each question, write either:

- a number or formula
- one of: True, False, or NEI (Not Enough Information)
- a brief definition or explanation (one sentence)

(a) (3 points) A monopolist faces demand with elasticity $\varepsilon = -2$. The Lerner index is:

(a) _____

(b) (3 points) Write the Berry inversion formula that relates market shares to mean utilities in the logit model.

(b) _____

(c) (3 points) True, False, or NEI: Adding demographic interactions to a logit model fully solves the IIA problem.

(c) _____

(d) (3 points) What is the economic interpretation of the price coefficient α in the logit demand model?

(d) _____

(e) (3 points) True, False, or NEI: If marginal cost increases by \$1, a monopolist with linear demand will raise price by exactly \$1.

(e) _____

(f) (3 points) True, False, or NEI: Under a two-part tariff with homogeneous consumers, the optimal per-unit price equals marginal cost.

(f) _____

(g) (3 points) Name one reason why the “BLP instruments” (characteristics of other products) help identify the price coefficient.

(g) _____

(h) (3 points) True, False, or NEI: In a self-selection problem, the binding incentive compatibility constraint is that the low type must not prefer the high-type product.

(h) _____

(i) (3 points) What is “versioning” in the context of self-selection?

(i) _____

(j) (3 points) True, False, or NEI: Consumer surplus in the logit model can be computed using the log-sum formula: $CS = \frac{1}{|\alpha|} \ln(1 + \sum_j \exp(v_j))$.

(j) _____

2. Demand Estimation (30 points)

2. Consider a market with 3 differentiated products. The logit demand model is:

$$u_{ijt} = \delta_{jt} + \alpha p_{jt} + \varepsilon_{ijt}$$

where $\alpha = -0.5$ is the price coefficient.

The following data are observed:

Product	Price (p_j)	Mean Utility (δ_{jt})
1	\$20	4.0
2	\$15	3.0
3	\$10	2.0

- (a) (8 points) Compute $v_j = \delta_{jt} + \alpha p_{jt}$ for each product. Then compute the market shares using:

$$s_{jt} = \frac{\exp(v_j)}{1 + \sum_k \exp(v_k)}$$

- (b) (6 points) Compute the own-price elasticity for each product using $\eta_{jj} = \alpha p_j(1 - s_j)$. Which product has the most elastic demand?

- (c) (6 points) A researcher estimates demand using OLS (regressing $\ln(s_{jt}) - \ln(s_{0t})$ on price and product characteristics). Explain why this leads to biased estimates of α . What is the direction of the bias?

- (d) (10 points) Using the log-sum formula, compute consumer surplus per consumer. If product 3 is removed from the market, what is the change in consumer surplus?

$$CS = \frac{1}{|\alpha|} \ln \left(1 + \sum_j \exp(v_j) \right)$$

3. Two-Part Tariffs (20 points)

3. A gym charges a monthly membership fee F and a per-visit price p . Marginal cost per visit is $c = 2$.

Part A: Homogeneous consumers

There are 100 identical consumers, each with demand $q = 20 - 2p$ for gym visits per month.

- (a) (5 points) Write down the gym's profit as a function of F and p . (Hint: What is consumer surplus at price p ? What is the maximum F the gym can charge?)

- (b) (5 points) Show that the profit-maximizing per-visit price is $p^* = c = 2$. (Hint: Take the derivative of profit with respect to p and set it equal to zero.)

Part B: Heterogeneous consumers

Now suppose there are 50 "heavy users" with demand $q_H = 10 - p$ and 100 "light users" with demand $q_L = 5 - p$. Marginal cost is $c = 0$.

- (c) (10 points) The gym is considering two pricing strategies:

- (a) $F = 50, p = 0$
- (b) $F = 12.5, p = 0$

Calculate profit under each strategy. Which is better and why?

4. Self-Selection (20 points)

4. A software company sells two versions: Pro (P) and Basic (B). There are two types of users: businesses with high willingness to pay and students with low willingness to pay.

Consumer Type	Willingness to Pay	
	Pro Version	Basic Version
Business (40 users)	\$400	\$150
Student (60 users)	\$120	\$100

Marginal cost is \$30 for both versions.

- (a) (4 points) If the company could perfectly identify consumer types, what prices would it charge for each version to each type? What is total profit?
- (b) (4 points) The company cannot identify types but offers both versions. Write down the incentive compatibility (IC) constraint that ensures businesses buy Pro rather than Basic.
- (c) (6 points) Suppose the company sets $P_P = \$380$ and $P_B = \$100$. Explain why these prices are not optimal. Which constraint is violated?
- (d) (6 points) Find the profit-maximizing prices for Pro and Basic under self-selection. (Hint: Which IC constraint binds? Set it as an equality and solve.)